# Factors Associated with Longevity in Companion Dogs: Initial Findings from the Dog Aging Project

## Supplementary Materials

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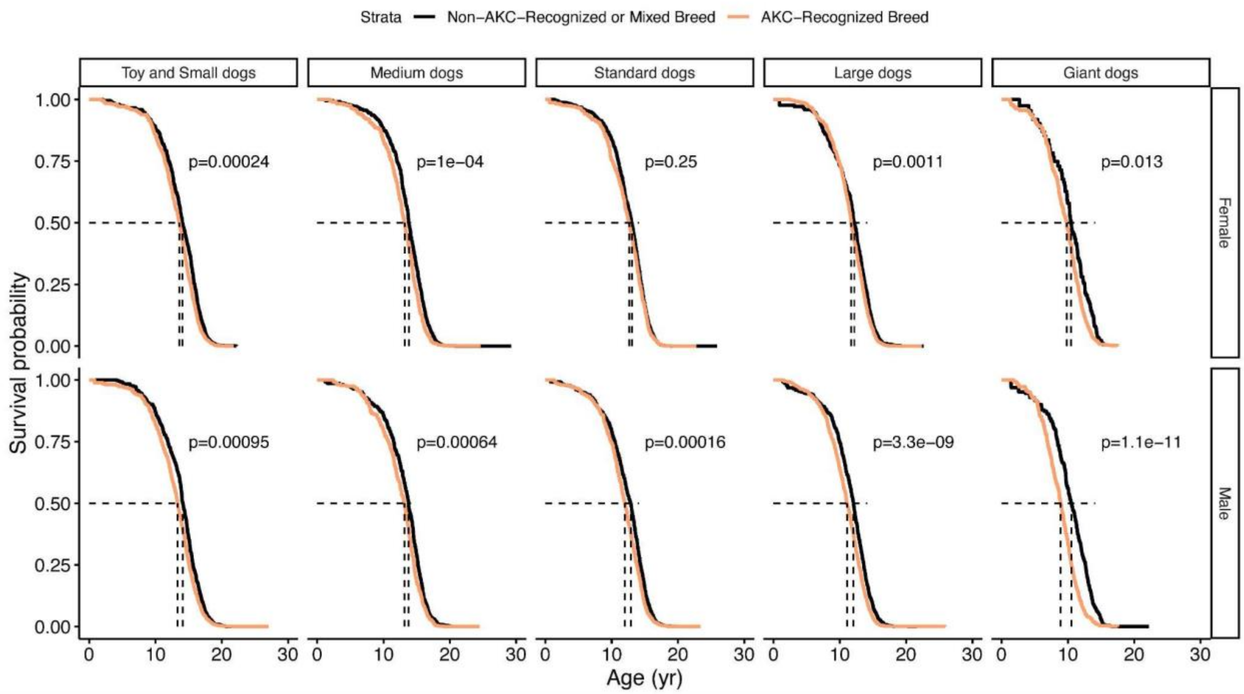
**Supplemental Table S1. Cohort Baseline Characteristics and Mortality Statistics by Sex, Breed Status, and Size Class.**

| Subgroup | N | Total dog-years | Deaths | Crude mortality rate | Age at baseline* | Years of follow-up* | Age of death* |
|---|---|---|---|---|---|---|---|
| Female Non-AKC-Recognized or Mixed Breed Toy and Small dogs | 2177 | 4916 | 549 | 111.7 per 1000 dog-years | 7.5 (3.89-11.15) [0-20.78] | 2.08 (1.05-3.72) [0.01-4.84] | 14.77 (12.72-16.32) [2.06-22.18] |
| Female Non-AKC-Recognized or Mixed Breed Medium dogs | 2816 | 6425 | 664 | 103.3 per 1000 dog-years | 6.16 (2.94-10.45) [0-25.54] | 2.1 (1.05-3.64) [0-4.74] | 14.4 (12.77-15.96) [1.24-21.73] |
| Female Non-AKC-Recognized or Mixed Breed Standard dogs | 4191 | 9282 | 1082 | 116.6 per 1000 dog-years | 6.21 (3-10) [0-23.49] | 2.07 (1.03-3.47) [0-4.69] | 13.69 (11.68-15.02) [1.42-25.87] |
| Female Non-AKC-Recognized or Mixed Breed Large dogs | 1264 | 2733 | 419 | 153.3 per 1000 dog-years | 7.48 (4-10.24) [0.15-22] | 2.06 (1.03-3.2) [0-4.72] | 13.01 (11.17-14.36) [0.9-22.64] |
| Female Non-AKC-Recognized or Mixed Breed Giant dogs | 292 | 576 | 101 | 175.3 per 1000 dog-years | 5.87 (2.99-9.35) [0.21-14] | 1.97 (1-3.05) [0.01-4.73] | 11.49 (9.87-13.37) [2.68-15.36] |
| Female AKC-Recognized Breed Toy and Small dogs | 2208 | 4761 | 757 | 159 per 1000 dog-years | 8.34 (3.78-12.13) [0.07-20.87] | 2.05 (1.03-3.31) [0-4.72] | 14.74 (12.94-16.21) [1.85-21.32] |
| Female AKC-Recognized Breed Medium dogs | 1440 | 3085 | 368 | 119.3 per 1000 dog-years | 6.18 (2.61-10.64) [0.18-23.49] | 2.04 (1.03-3.2) [0-5] | 14.03 (12.17-15.37) [1.87-20.8] |
| Female AKC-Recognized Breed Standard dogs | 1881 | 4126 | 472 | 114.4 per 1000 dog-years | 5.63 (2.5-9.87) [0.1-20.49] | 2.07 (1.03-3.37) [0-4.64] | 13.4 (11.41-14.93) [0.65-22.78] |
| Female AKC-Recognized Breed Large dogs | 3484 | 7691 | 1060 | 137.8 per 1000 dog-years | 6 (2.65-9.72) [0.05-20.8] | 2.07 (1.04-3.37) [0-4.71] | 12.59 (10.85-14.01) [2.54-22.36] |
| Female AKC-Recognized Breed Giant dogs | 781 | 1626 | 260 | 159.9 per 1000 dog-years | 4.8 (2.43-8) [0.14-15.61] | 2.01 (1.03-3.08) [0-4.7] | 10.61 (8.6-12.12) [1.3-17.47] |
| Male Non-AKC-Recognized or Mixed Breed Toy and Small dogs | 2126 | 4748 | 538 | 113.3 per 1000 dog-years | 7.69 (3.77-11.26) [0-24.41] | 2.08 (1.03-3.55) [0-5.01] | 14.74 (12.87-16.43) [4.15-22.63] |
| Male Non-AKC-Recognized or Mixed Breed Medium dogs | 1933 | 4324 | 525 | 121.4 per 1000 dog-years | 7 (3.13-11) [0-22.68] | 2.08 (1.05-3.45) [0-4.65] | 14.5 (12.65-15.91) [1.19-20.59] |
| Male Non-AKC-Recognized or Mixed Breed Standard dogs | 3194 | 7043 | 845 | 120 per 1000 dog-years | 6 (2.99-10) [0-18] | 2.08 (1.03-3.35) [0-4.69] | 13.55 (11.38-14.95) [0.81-19.82] |

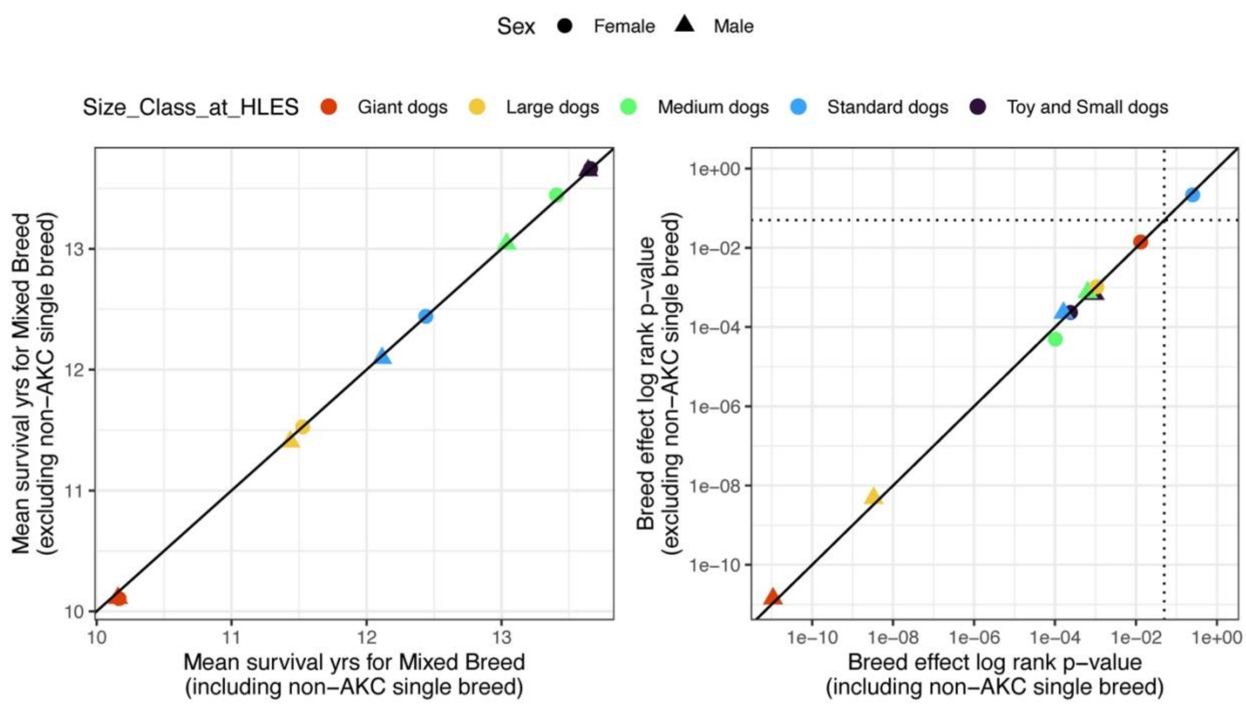
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Male Non-AKC-Recognized or Mixed Breed Large dogs | 2142 | 4638 | 655 | 141.2 per 1000 dog-years | 6.71 (3.45-10) [0-20.59] | 2.06 (1.03-3.29) [0.01-4.64] | 12.7 (10.89-14.07) [1.39-20.29] |
| Male Non-AKC-Recognized or Mixed Breed Giant dogs | 827 | 1708 | 287 | 168 per 1000 dog-years | 6 (3.12-9.25) [0.23-21.18] | 2.03 (1.01-3.07) [0-5] | 11.84 (9.68-13.33) [1.4-22.19] |
| Male AKC-Recognized Breed Toy and Small dogs | 2496 | 5448 | 890 | 163.4 per 1000 dog-years | 8.75 (3.79-12.42) [0.19-23.31] | 2.05 (1.03-3.37) [0-4.77] | 14.7 (13.04-16.21) [0.81-22.82] |
| Male AKC-Recognized Breed Medium dogs | 1454 | 3200 | 396 | 123.7 per 1000 dog-years | 6 (2.68-10.43) [0.18-20.38] | 2.06 (1.05-3.27) [0.02-4.71] | 13.99 (11.93-15.58) [2.39-24.45] |
| Male AKC-Recognized Breed Standard dogs | 2018 | 4457 | 550 | 123.4 per 1000 dog-years | 5.55 (2.47-9.42) [0.1-21.67] | 2.08 (1.03-3.41) [0-4.66] | 12.86 (10.76-14.42) [1.35-18.56] |
| Male AKC-Recognized Breed Large dogs | 3494 | 7707 | 1040 | 134.9 per 1000 dog-years | 5.29 (2.66-8.93) [0-24.83] | 2.06 (1.04-3.29) [0-4.71] | 12.03 (9.81-13.52) [1.48-25.85] |
| Male AKC-Recognized Breed Giant dogs | 829 | 1736 | 281 | 161.9 per 1000 dog-years | 4 (2-7.01) [0.17-15.71] | 2.05 (1.03-3.06) [0-4.7] | 9.82 (7.57-11.26) [1.82-17.51] |

*Median (IQR) [Range]

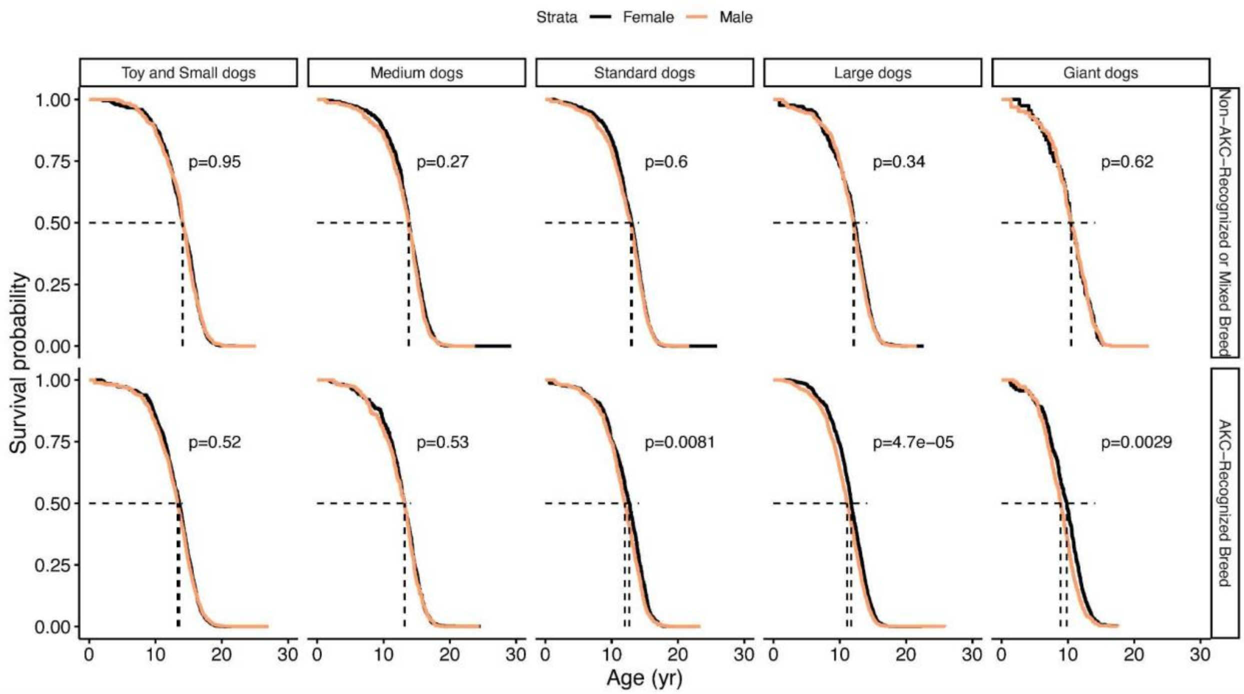
**Supplementary Table S2 (provided as separate Excel file):** Survey Wide Association Study variables with their p-values and hazard ratios, for main analysis and sensitivity analyses.



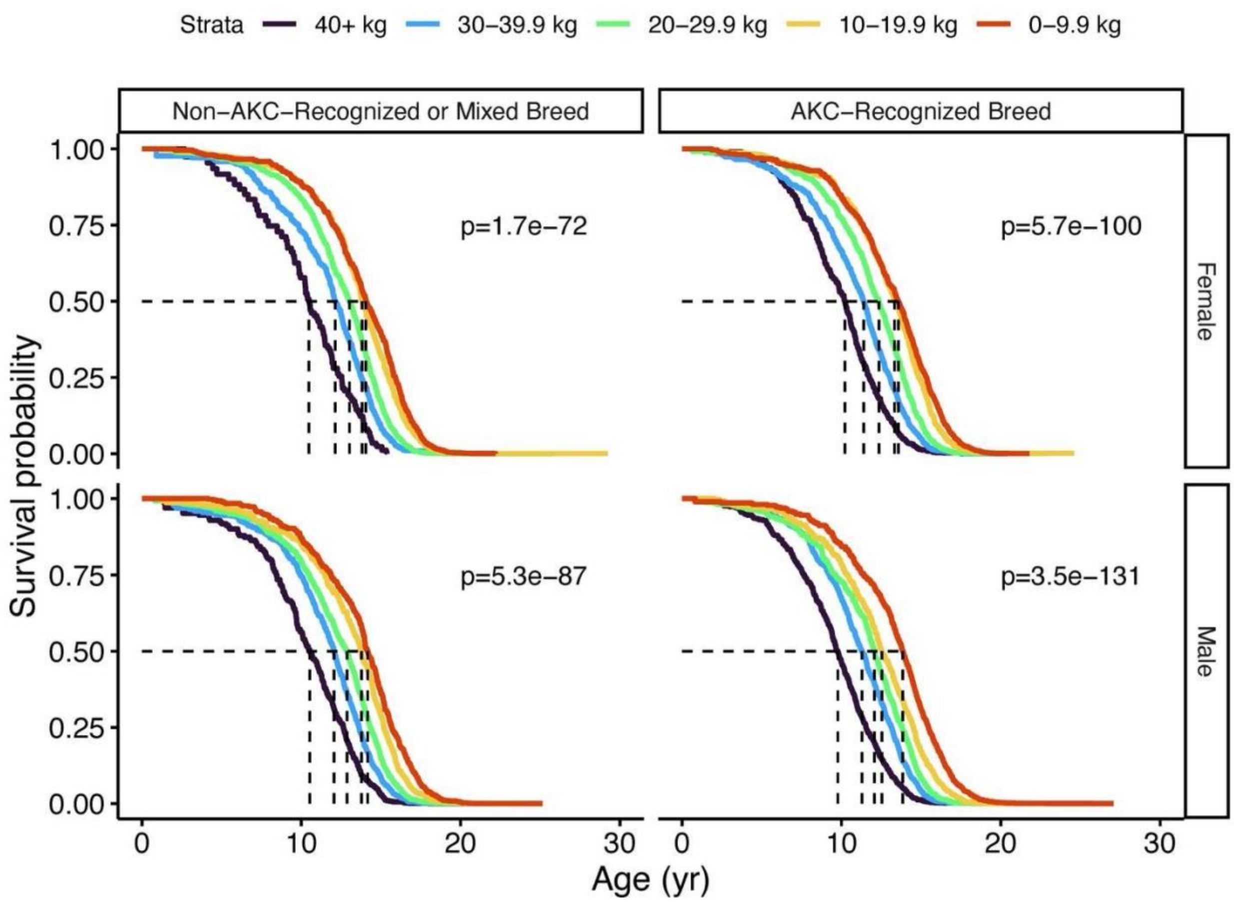

**Supplemental Figure S1.** Kaplan-Meier survival curves comparing breed classes (with log-rank p-values), stratified by size class and sex.



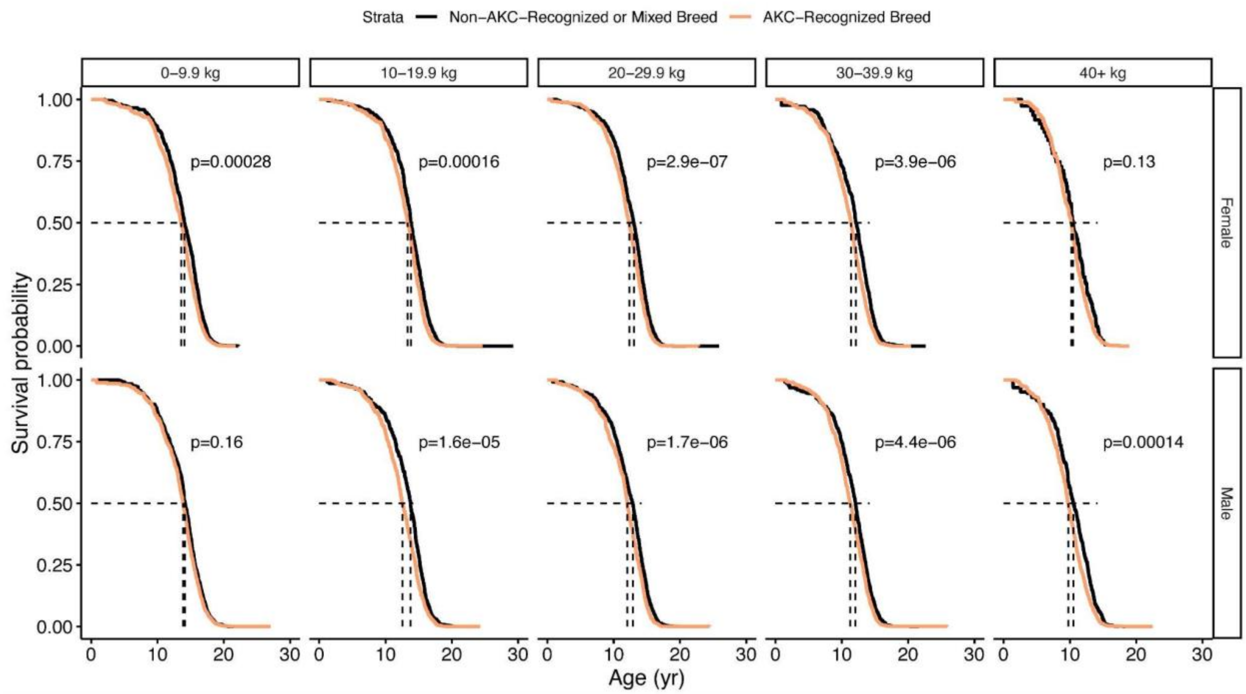

**Supplemental Figure S2.** Comparison of mean lifespan and log-rank p-values for breed classes when including (main analysis) or excluding (sensitivity analysis) non-AKC single breed dogs from the mixed breed category.



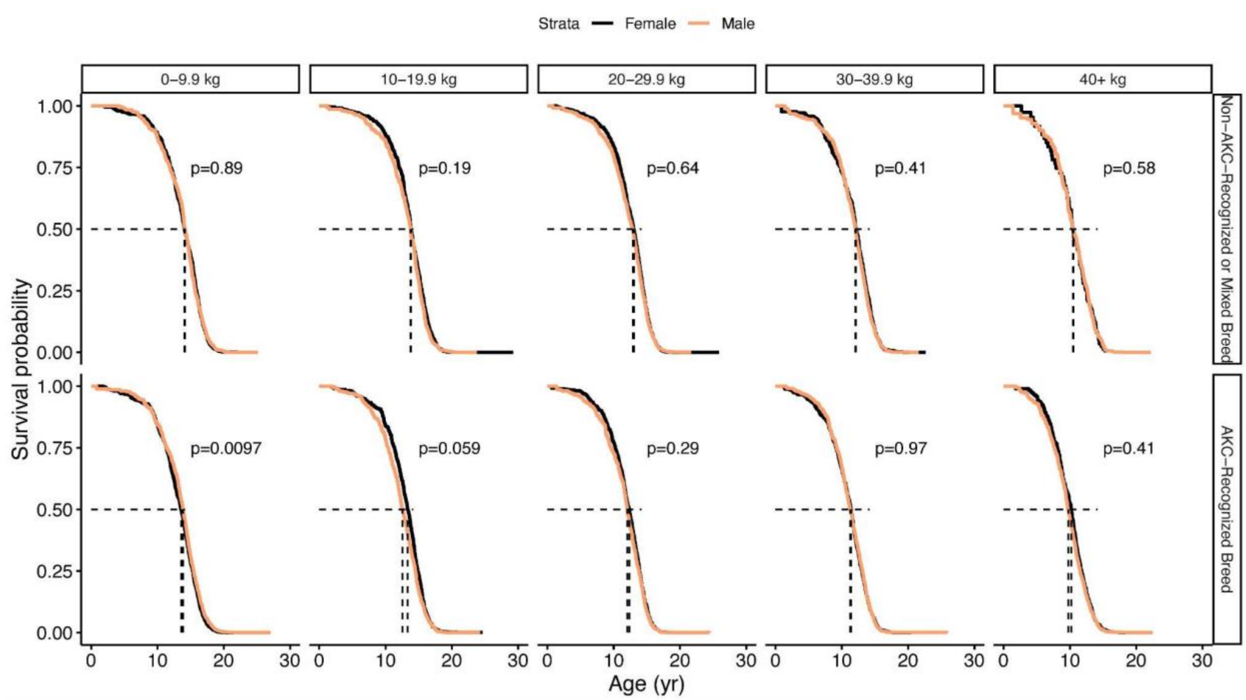

**Supplemental Figure S3.** Kaplan-Meier survival curves comparing sexes (with log-rank p-values), stratified by size class and breed class.



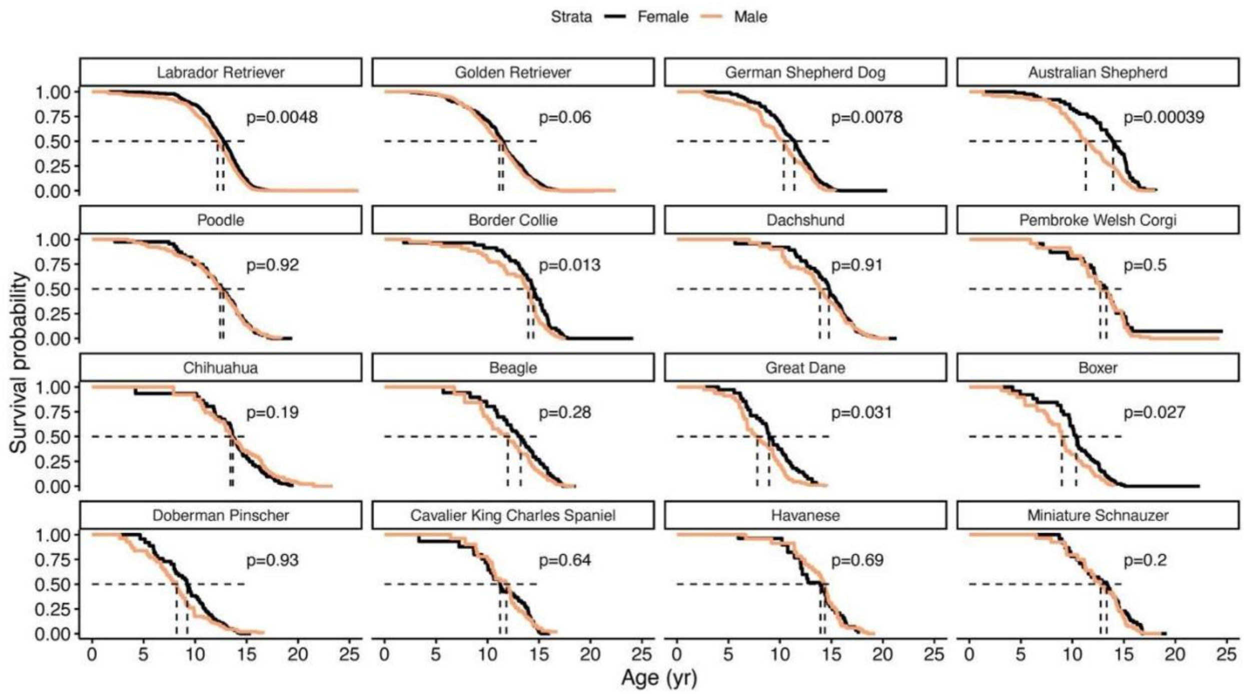

**Supplemental Figure S4.** Sensitivity analysis - Kaplan-Meier Survival Curves for different owner-reported weight categories instead of size categories that account for breed (for single breed dogs), stratified by breed status and sex.



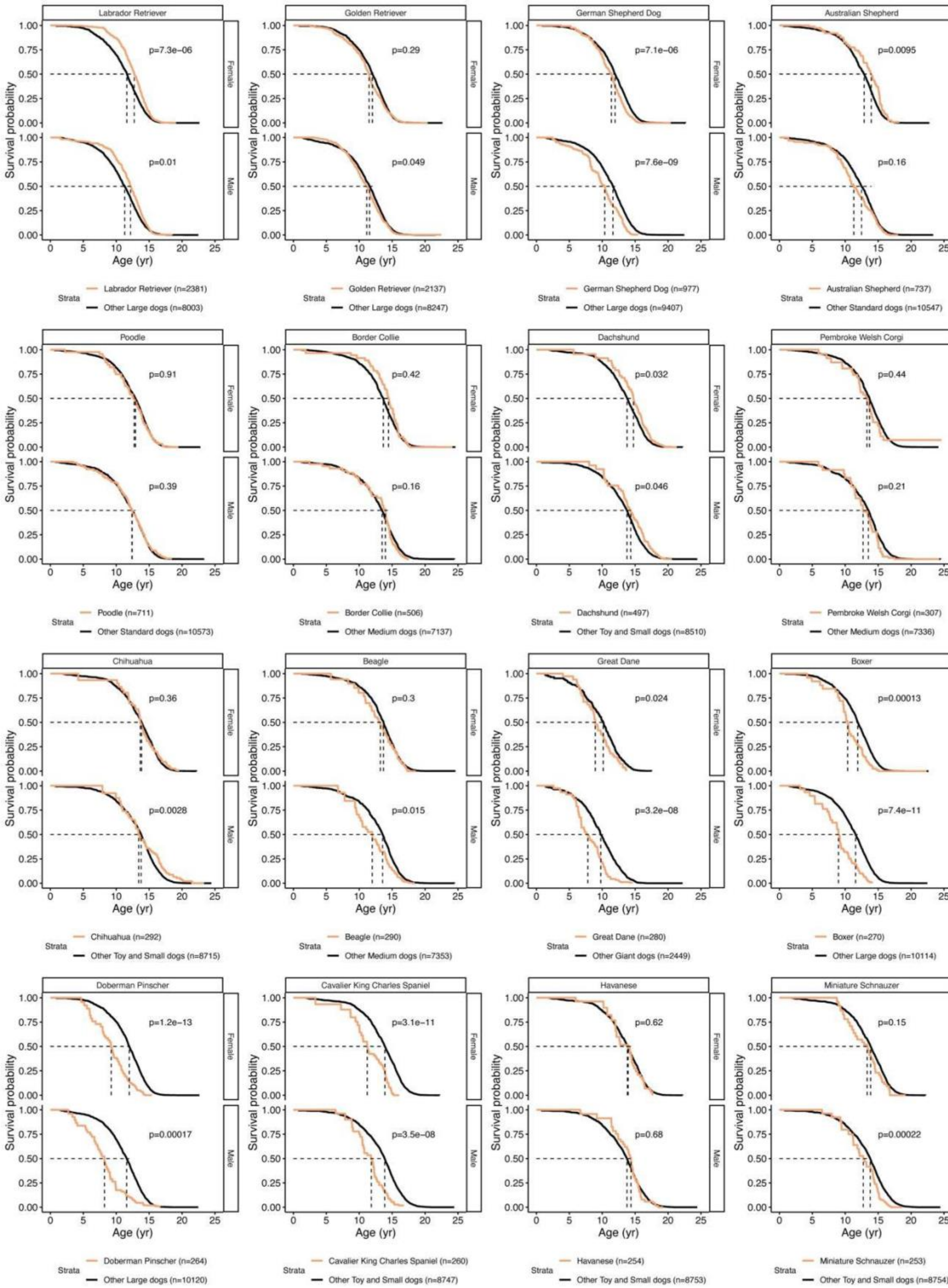

**Supplemental Figure S5.** Sensitivity analysis - Kaplan-Meier Survival Curves comparing breed classes (with log-rank p-values), stratified by owner-reported weight categories and sex.



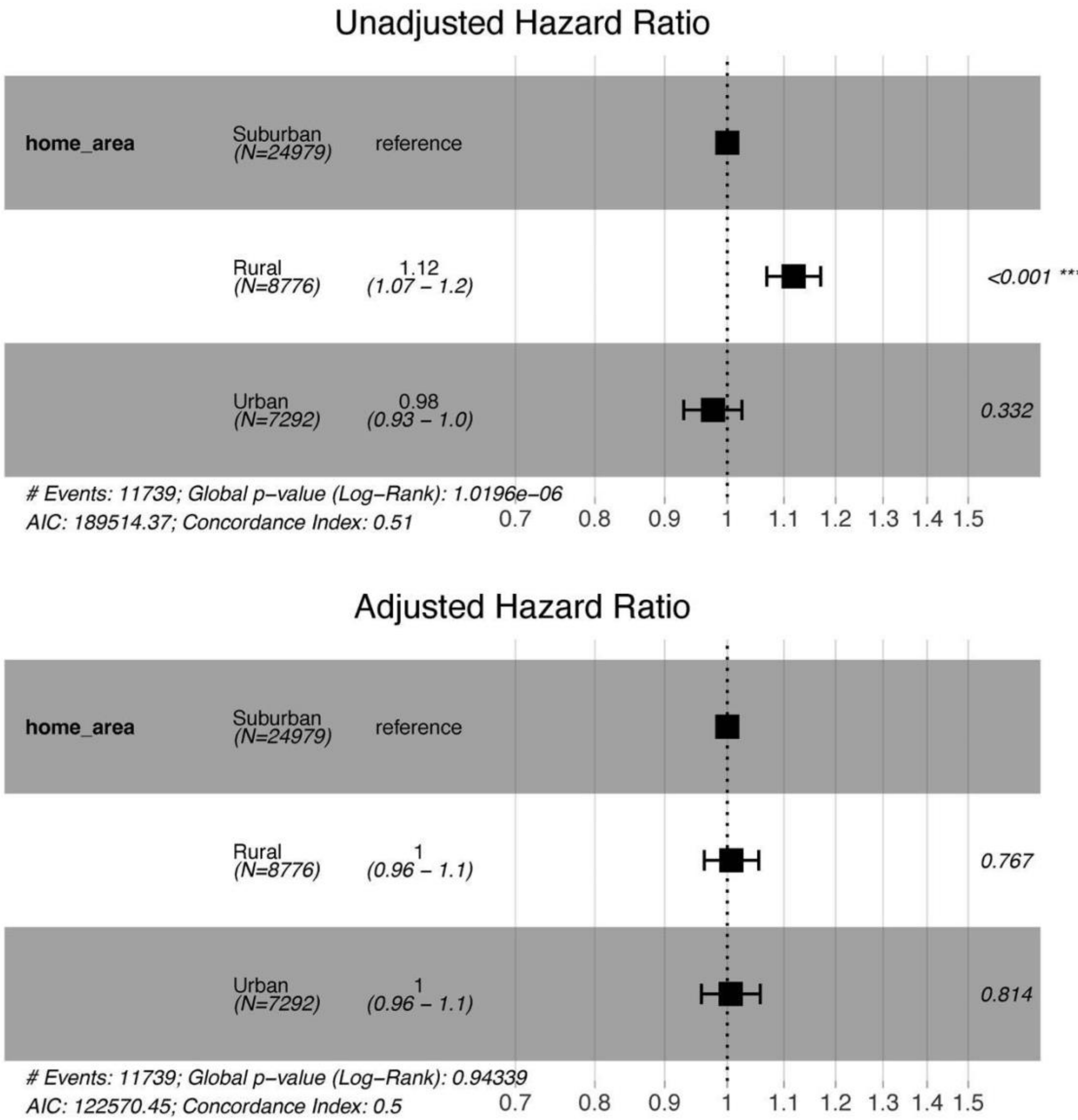

**Supplemental Figure S6.** Sensitivity analysis - Kaplan-Meier survival curves comparing sexes (with log-rank p-values), stratified by owner-reported weight categories and breed class.



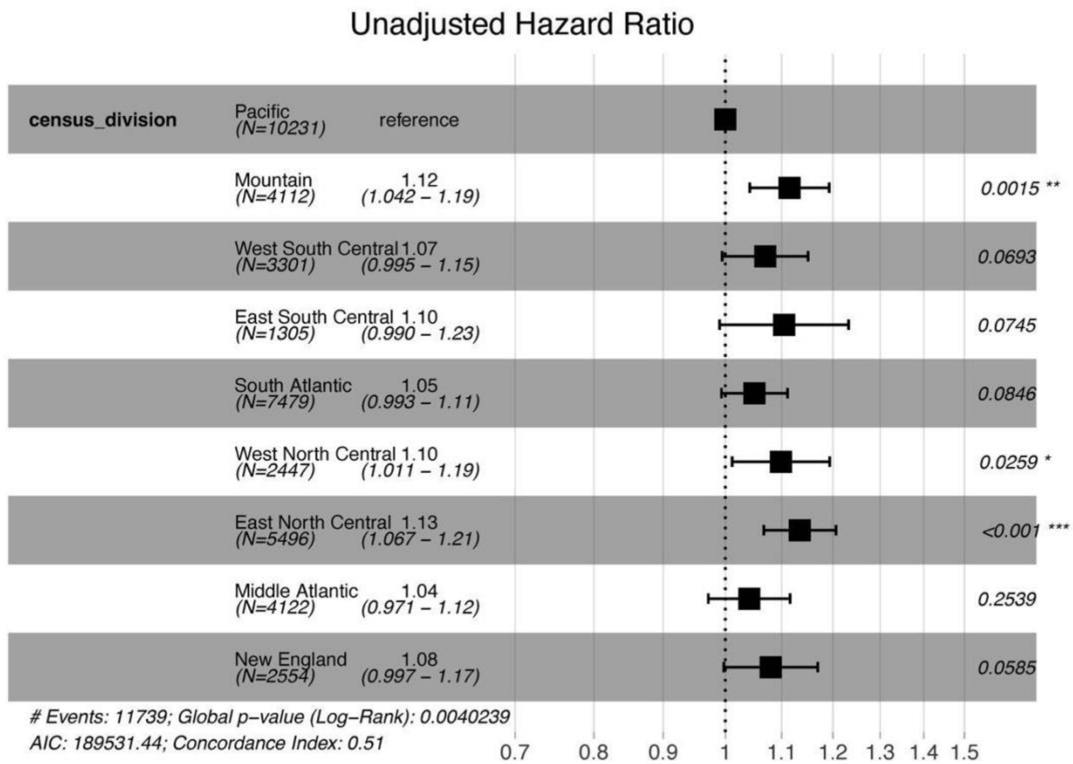

**Supplemental Figure S7.** Kaplan-Meier survival curves comparing sexes (with log-rank p-values) for 16 most common single breed dogs in cohort.

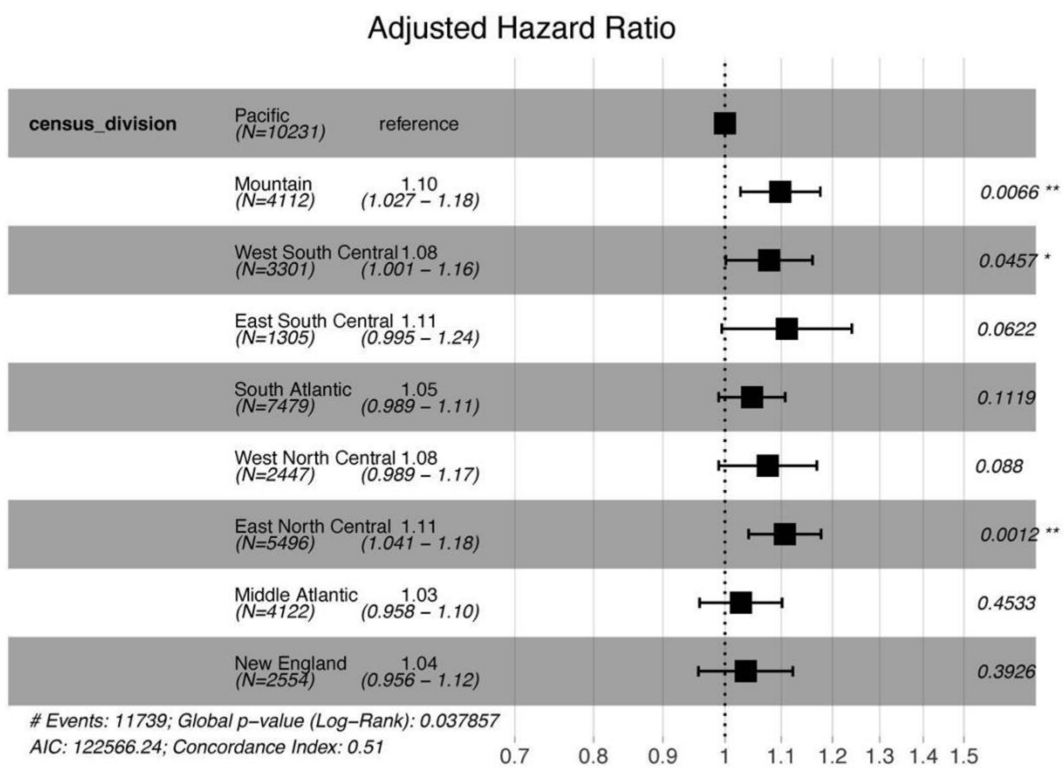


**Supplemental Figure S8.** Kaplan-Meier survival curves comparing each breed and sex with its respective size class/sex stratum (with log-rank p-values) for 16 most common single breed dogs in cohort.

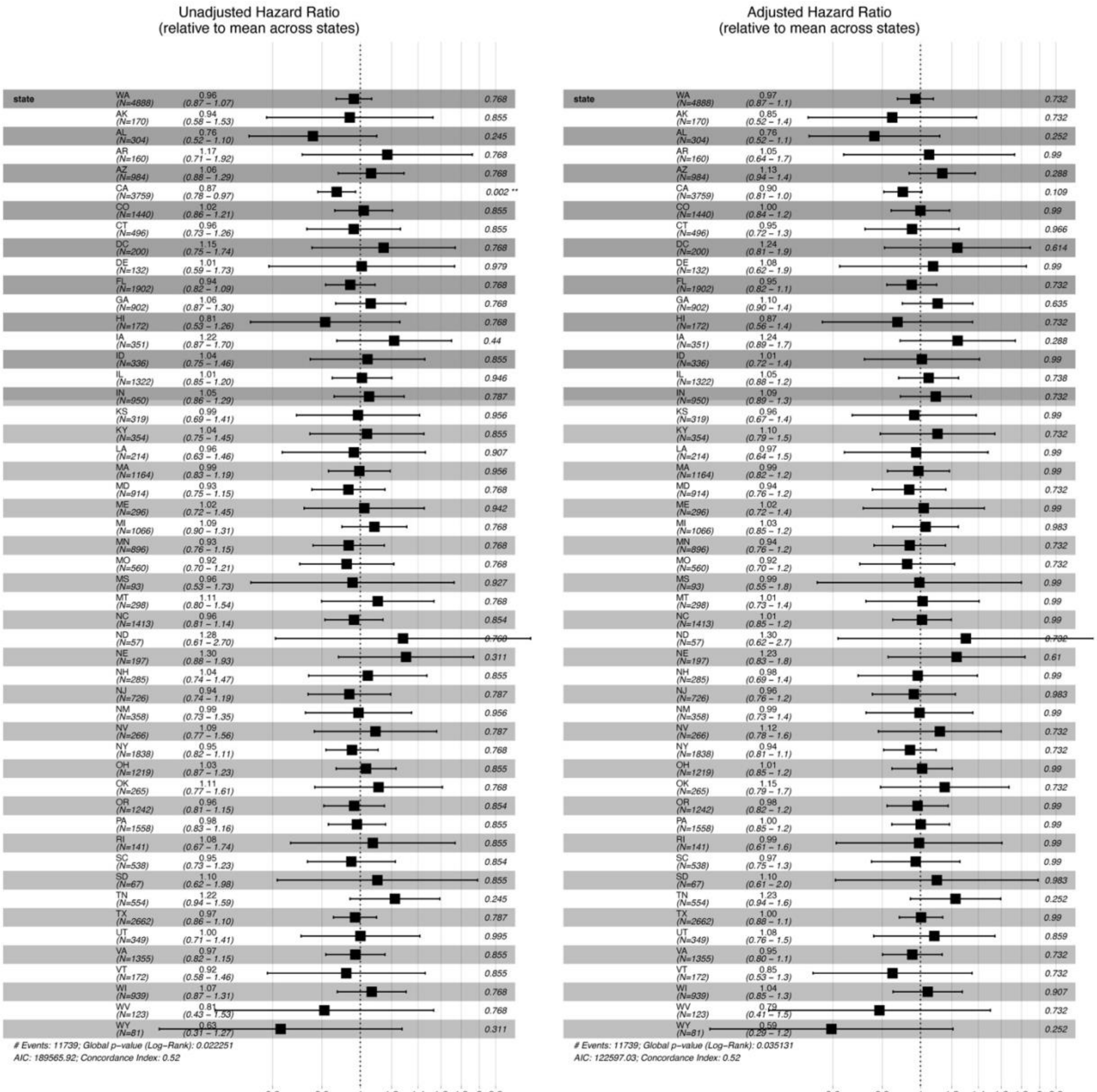



**Supplemental Figure S9.** Hazard ratios for different home area types (rural, suburban, urban) unadjusted and adjusted for demographic factors (size class, breed class, sex).

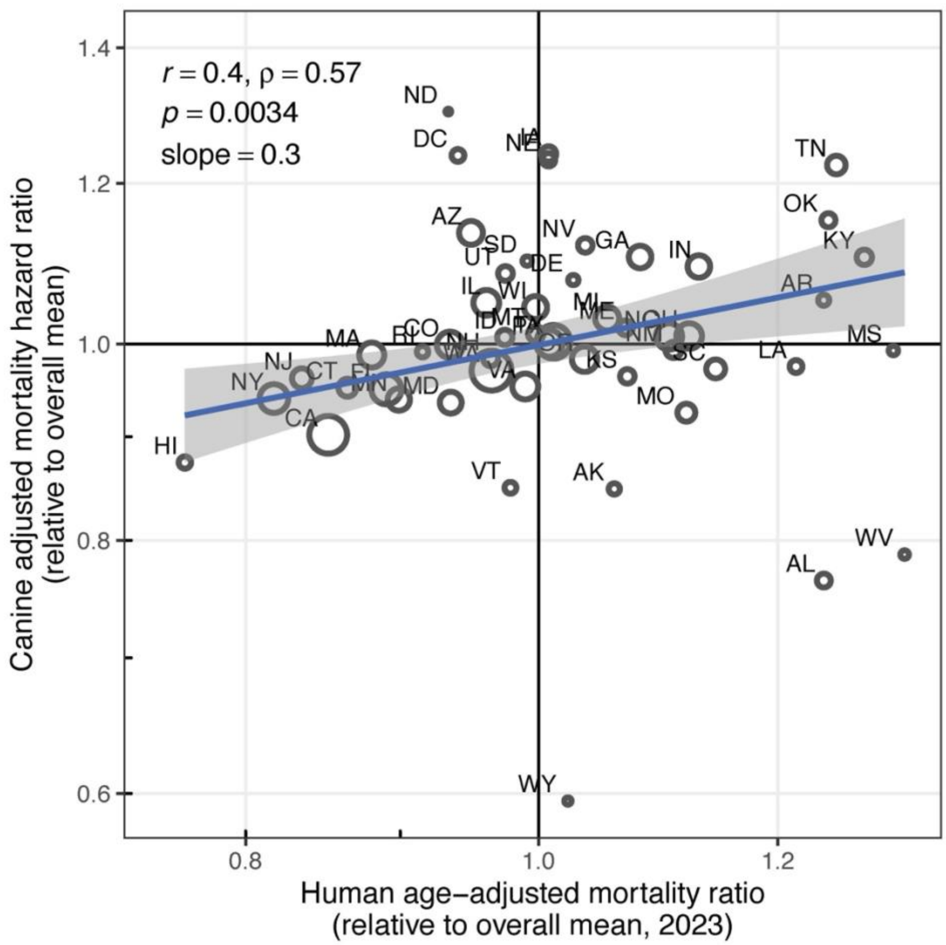


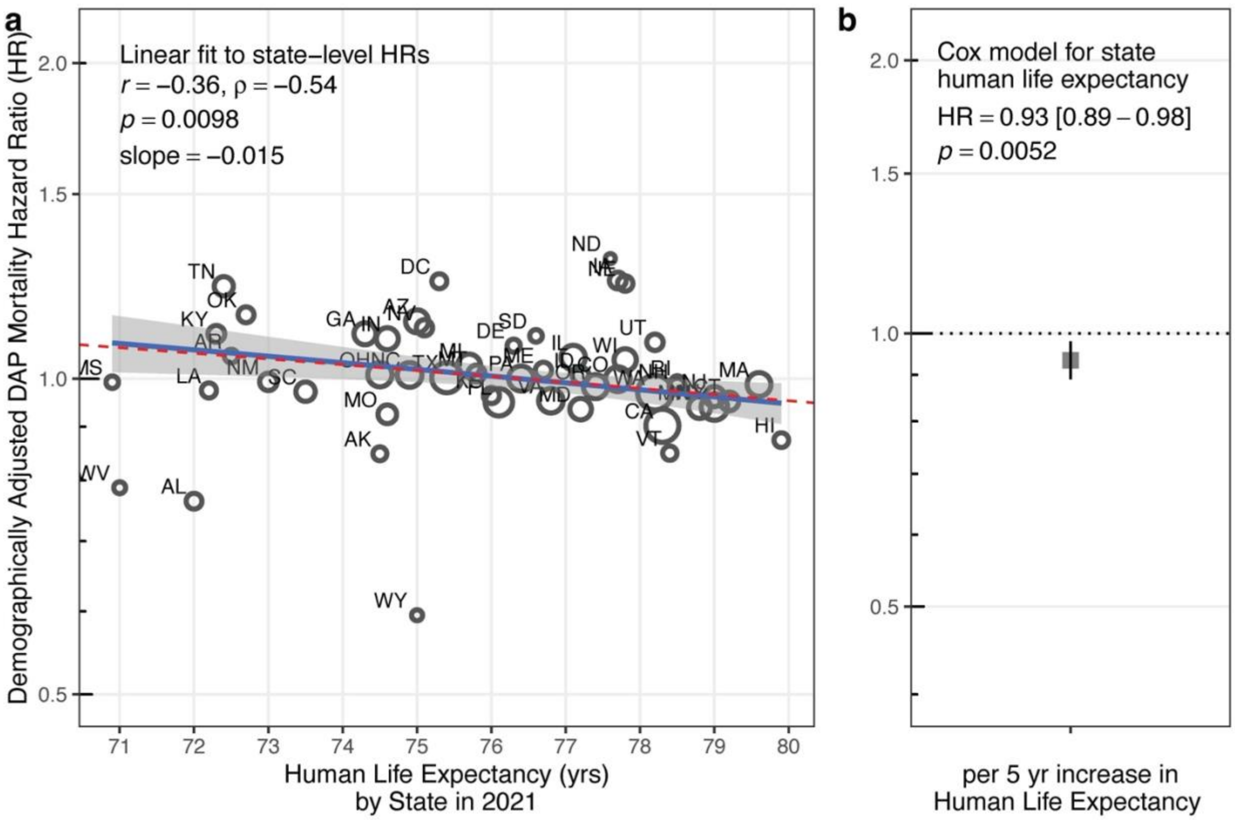

**Supplemental Figure S10.** Hazard ratios for different census divisions unadjusted and adjusted for demographic factors (size class, breed class, sex).



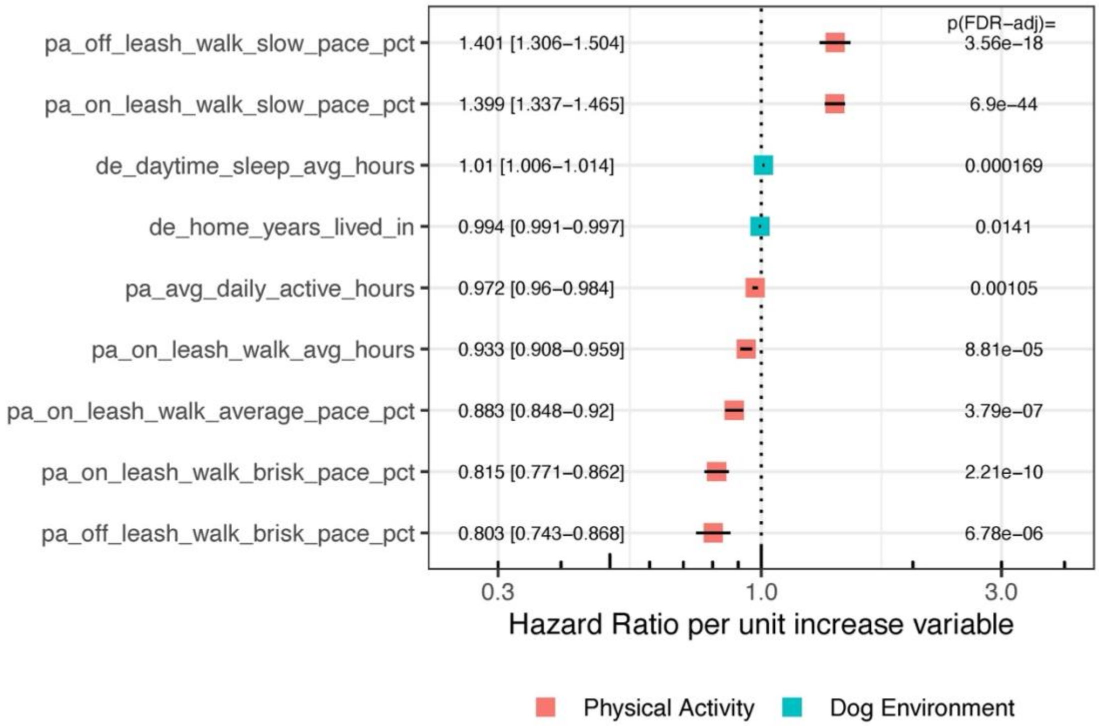
**Supplemental Figure S11.** Hazard ratios for different states and the District of Columbia unadjusted and adjusted for demographic factors (size class, breed class, sex).




**Supplemental Figure S12** (same as **Figure 2**, with human state-level age-adjusted mortality rates restricted to 2023). Comparison of demographically adjusted dog hazard ratios (HR) by state (see **Supplemental Figure S11**) with state-level age-adjusted mortality ratios for 2023. Both quantities are expressed relative to the count-weighted overall mean across states: the dog HR as a ratio to the mean dog hazard, and the human value as each state's age-adjusted mortality rate (AAMR) divided by the count-weighted mean AAMR across states. Because both axes are centered on the overall mean rather than on a reference state, values above or below 1 indicate higher or lower mortality than the average state.  Circle sizes denote regression weights, based on inverse variance of the dog HR; human AAMR standard errors (<1%) are much smaller and would contribute negligibly to weighting. The thick solid line and shaded region show linear trend of log(HR) vs. log(mortality ratio) and its confidence band, from weighted linear regression. Also shown are Pearson (r) and Spearman (ρ) correlation coefficients, the *p*-value, and slope.

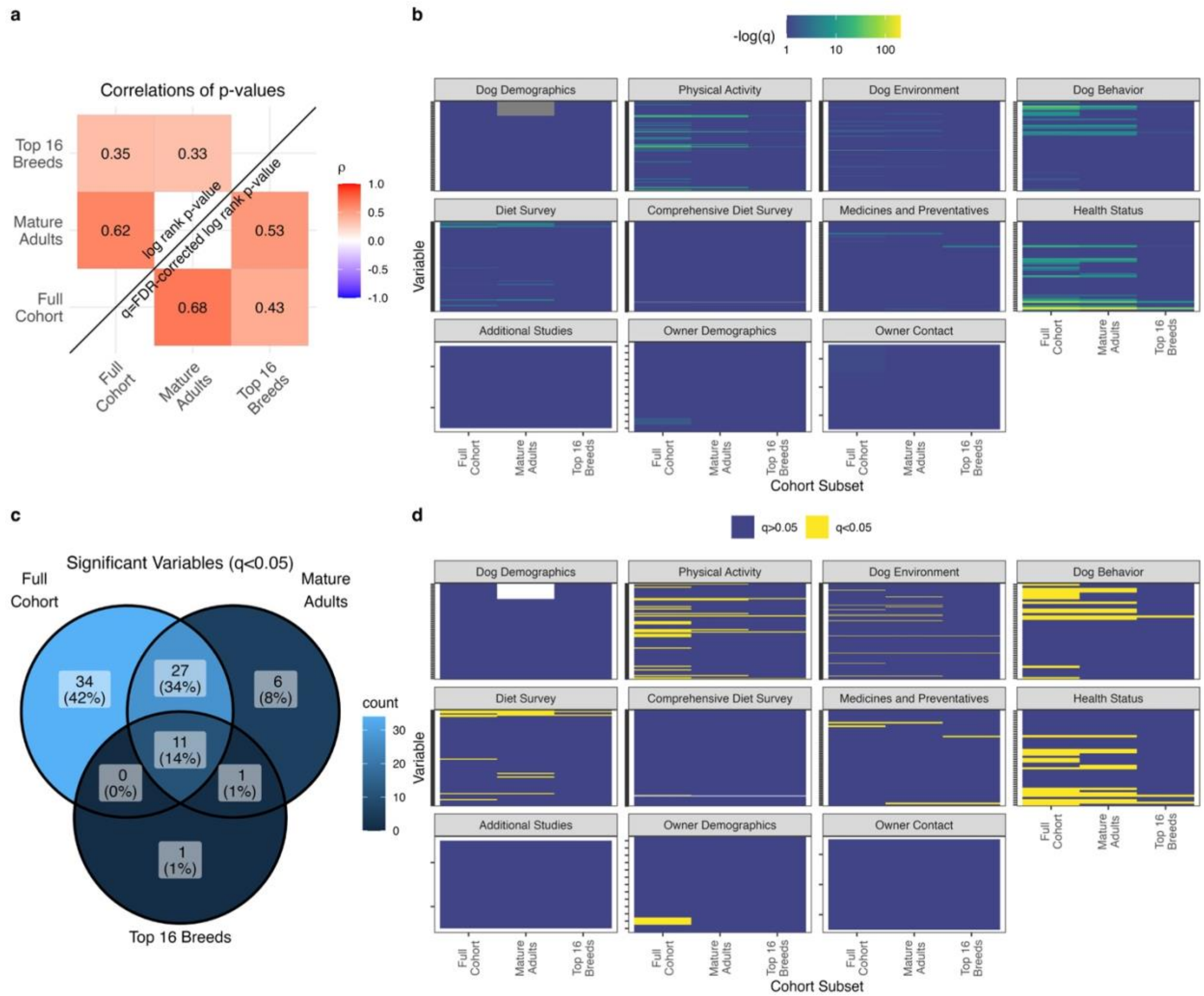



**Supplemental Figure S13. a:** Comparison of demographically adjusted dog hazard ratios (HR) by state (see **Supplemental Figure S11**) with state-level human life expectancy at birth for 2021 (LE). The thick solid line and shaded region show linear trend and confidence band, based on linear regression weighted by inverse variance of the dog hazard ratio. The dashed line shows a one-to-one relationship where a change in dog hazard ratio leads to an equal (negative) relative change in human life expectancy (HR – 1) = (1 – LE /76.4), where 76.4 is the overall US 2021 LE. **b:** demographically adjusted dog HR for a 5 year increase in LE based on Cox proportional hazards model using state-level LE as a continuous variable.




**Supplemental Figure S14.** Forest plots of hazard ratios and FDR-corrected p-values for numeric HLES variables from SWAS. Each hazard ratio (and confidence interval) corresponds to a unit increase of the numeric variable (pct=change from 0 to 100% of the time; years_lived_in=increase in 1 year; hours=increase in 1 hour).

**Supplemental Figure S15 (provided as separate PDF).** Forest plots of hazard ratios and FDR-corrected p-values for categorical HLES variables from SWAS. Each panel shows the variable abbreviation and corresponding survey question in the HLES. Each hazard ratio (and confidence interval) corresponds to the increase or decrease relative to the reference category, which is defined by the category with the most frequent response.




**Supplemental Figure S16.** Sensitivity analysis for Survey-Wide Association Study analysis using cohort subsets of mature adult dogs only and top 16 (by DAP dog number) single breeds only. **a:** Spearman correlation of log rank p-values, unadjusted (upper left) and adjusted (lower right) using False Discovery Rate approach (q-values). **b:** Heatmap of -log10(q)-values across cohort subsets, arranged by survey module. **c:** Venn diagram of significant variables (q < 0.05) across cohort subsets. **d:** Heatmap of significant q-values across cohort subsets, arranged by survey module.